MUHAMMAD HASSAN

GAME DEVELOPER | GAME DESIGNER

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PROFESSIONAL SUMMARY

Unreal Engine developer with 5+ years of experience building gameplay systems, multiplayer features, tools, and production interactive systems in C++ and Blueprints. Hands-on with replication, RPCs, EOS, session flows, multiplayer UI, debugging, profiling, and optimization. Translates gameplay requirements into working features and owns implementation through iteration and release support.

CORE SKILLS

Unreal Engine: UE5.x, C++, Blueprints, UMG, editor tools/plugins, Unreal source builds

Multiplayer: replication, RPCs, EOS, session creation/discovery/join/leave, server browsing, synchronized gameplay, network debugging

Gameplay/technical design: gameplay systems implementation, modular architecture, prototyping, iteration, player-facing flows, validation

Animation: Animation Blueprints, locomotion, motion matching, MetaHuman integration, runtime character customization

Performance and delivery: CPU/GPU profiling, memory and asset streaming, level streaming, packaging, optimization, logging

Tools: custom plugins, Blueprint-callable tools, GitHub, Perforce, documentation, release support

PROFESSIONAL EXPERIENCE

Moshpit Studio — Game Developer | Jul 2021–Present

- Built maintainable gameplay foundations, measured by modular systems, mechanics, animation pipelines, and C++/Blueprint architecture, by implementing reusable feature structures.
- Enabled synchronized browser play, measured by replicated player interactions in Pixel Streaming experiences, by delivering networked multiplayer features.
- Supported two target platforms, measured by PC and Android release workflows, by building UMG, plugins, Blueprint tools, and technical documentation.
- Reduced repeated designer setup, measured by tool panels being generated and bound from data, by implementing data-driven editor UI generation.
- Reduced manual cinematic setup, measured by shot plans producing CineCamera actors, animation, and cuts, by developing an AI-driven cinematics system.
- Enabled repeatable spatial capture, measured by generated grids/routes, mesh bounds, validation logs, and GPU readback, by building autonomous navigation and capture tools.
- Supported release readiness, measured by profiling, debugging, optimization, packaging, and streaming workflows, by resolving gameplay and asset-delivery issues.

Skylla Studio (Greece) — Multiplayer Developer | Apr 2025–Present

- Implemented EOS multiplayer flows, measured by session creation, discovery, and client join/leave behavior, by developing the game on an Unreal source build.
- Enabled synchronized networked actions, measured by replicated player interaction, by implementing gameplay state with replication and RPCs.
- Connected player flow to server behavior, measured by UMG menus reflecting session logic, by integrating screens with multiplayer systems.

Independent Game Developer — CerealKiller | Personal project

- Implemented an original UE 5.6 multiplayer game, measured by authoritative movement, validated dive, match phases, synchronized timers, and finish placement, by developing C++ gameplay and server rules.
- Configured two EOS lobby paths, measured by public capacity of 20 and party capacity of 10, by implementing login, invites, lobby state, discovery, and travel.

- Added multiplayer safeguards and recovery, measured by replicated match/player state, rate-limited chat, host/readiness validation, and migration logic, by building server-side systems.
- Built a reusable course hazard set, measured by 13 Blueprint bases plus seeded placement and day/night systems, by implementing hazard logic and chunk streaming.
- Delivered mobile-oriented player flows, measured by Jump/Dive controls, generated HUD/menu, appearance selection, and platform configurations, by integrating UI and Windows/Android support.
- Documented local verification and remaining gaps, measured by recorded builds/automation and explicitly listed live EOS/device checks, by maintaining implementation evidence.

SELECTED PROJECTS

Multiplayer Squads Game (Invasion) | Mar 2025–Present — Built EOS squad/session creation, server browsing, achievements, and replicated gameplay flows.

Virtual Concert Experience — UE5 Technical Systems | Aug 2022–Present — Built show-control, Nanite HISM crowds, camera/cinematic tooling, VR camera paths, runtime clothing changes, Pixel Streaming, and streaming/memory optimization.

UnrealEngineTwin | May 2025–Present — Implemented EQS navigation and visibility-aware scanning, UE5.5 capture/export using RDG compute and GPU readback, and Blueprint-callable tools with logging and validation.

EDUCATION

BS, Information Technology — Punjab University College of Information Technology (PUCIT) | 2018–2022